✎ Climate Policies Under Wealth Inequality

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Abstract

Context. Climate change mitigation is a paradigmatic collective-action problem in which individual incentives conflict with collective welfare. This problem is exacerbated by wealth inequality and heterogeneous risk perceptions, which shape both incentives and social learning dynamics.
Goal. The goal of this project is to reproduce the main qualitative and quantitative results of the evolutionary model proposed by Vasconcelos et al. (2014), which studies how inequality, risk, and homophily jointly affect cooperation in a collective-risk public goods game.
Method. We independently reimplemented the model as a finite-state Markov process. Payoffs were computed via exact multivariate hypergeometric sampling, and the stationary distribution of the induced evolutionary dynamics was obtained by solving the associated eigenvalue problem.
Results. Our implementation reproduces the original phase diagrams, stationary distributions, and group achievement curves across different risk and homophily regimes. We confirm the asymmetric impact of obstinate agents and the destabilizing role of homophily.
Conclusions. The reproduction confirms that inequality alone does not necessarily undermine cooperation. Instead, segregation in social learning is the dominant destabilizing mechanism, while persistent cooperation among poorer agents plays a stabilizing role.

1 Introduction

Climate change mitigation constitutes a collective action problem in which individual incentives conflict with collective welfare.

Vasconcelos et al. [1] introduced an evolutionary game-theoretic framework that models climate cooperation as a collective-risk threshold public goods game. Their analysis demonstrates that inequality alone does not necessarily undermine cooperation; instead, homophily in social learning and imitation plays a central destabilizing role.

The objective of this project is to replicate the main qualitative and quantitative results of the original paper using an independent implementation. Our replication relies on an exact computation of the stationary distribution of a finite-state Markov chain.

We focus on reproducing (i) the dependence of average group achievement on perceived risk, (ii) the structure of stationary distributions under varying homophily levels, and (iii) the asymmetric effects of obstinate agents across wealth classes.

2 Related Work

Milinski et al. [2] introduce the collective-risk social dilemma, in which groups must reach a contribution threshold to avoid a probabilistic collective loss. This framework was among the first to explicitly model catastrophic outcomes rather than smooth payoff degradation, making it particularly suitable for climate-related problems. While Milinski et al. focus on behavioral experiments and symmetric populations, Vasconcelos et al. extend this framework by incorporating wealth heterogeneity and evolutionary dynamics, allowing for an analysis of long-run population-level behavior under inequality.

Santos et al. [3] study the role of social diversity in public goods games and show that heterogeneity can promote cooperation by stabilizing mixed equilibria. Their work highlights that population structure and diversity can have counterintuitive stabilizing effects. However, their model does not include collective-risk thresholds or catastrophic losses. Vasconcelos et al. instead embeds heterogeneity into a collective-risk setting and by modeling imitation dynamics, connecting diversity effects to climate-relevant payoff structures.

Sigmund [4] provides a comprehensive theoretical treatment of evolutionary game dynamics in social dilemmas, emphasizing the role of imitation, mutation, and stochasticity. While this work offers a general methodological foundation, it does not address climate dilemmas or wealth inequality specifically. Vasconcelos et al. apply these general evolutionary principles to a concrete domain.

3 Model Definition

3.1 Population Structure

We consider a finite population of size Z subdivided into two wealth classes: Z_R rich individuals with endowment b_R and $Z_P = Z - Z_R$ poor individuals with endowment $b_P < b_R$. Each individual adopts one of two strategies, cooperation (C) or defection (D).

Groups of size N are formed uniformly at random without replacement.

3.2 Contributions and Threshold

Cooperators contribute a fixed fraction c of their endowment:

$$c_R = cb_R, \qquad c_P = cb_P.$$

Let j_R and j_P denote the number of rich and poor cooperators in a group. The group succeeds if total contributions exceed a fixed threshold:

$$j_R c_R + j_P c_P \geq Mc\bar{b},$$

where $\bar{b}$ is the average endowment in the population.

3.3 Payoffs

If the group fails, all individuals lose their remaining endowment with probability r. Using the Heaviside function $\Theta(\cdot)$, the expected payoff for defectors is

$$\Pi^D_{R/P}(j_R, j_P) = b_{R/P}\left[\Theta(\Delta) + (1-r)(1-\Theta(\Delta))\right],$$

where $\Delta = j_R c_R + j_P c_P - Mc\bar{b}$.

The payoff for cooperators is obtained by subtracting the contribution cost:

$$\Pi^C_{R/P}(j_R, j_P) = \Pi^D_{R/P}(j_R, j_P) - c_{R/P}.$$

4 Methods

4.1 Fitness Calculation

Fitness is defined as the expected payoff averaged over all possible group compositions, computed using multivariate hypergeometric sampling.

For example, the fitness of a rich cooperator in state (i_R, i_P) is:

$$f^C_R(i_R, i_P) = \sum_{j_R, j_P} \frac{\binom{i_R-1}{j_R}\binom{i_P}{j_P}\binom{Z-i_R-i_P}{N-1-j_R-j_P}}{\binom{Z-1}{N-1}} \Pi^C_R(j_R+1, j_P).$$

Analogous expressions apply for rich defectors and poor agents.

4.2 Evolutionary Dynamics

Strategy updates follow a pairwise comparison rule with Fermi imitation:

$$P(X \to Y) = \frac{1}{1 + e^{\beta(f_X - f_Y)}},$$

where β controls selection intensity.

Homophily is introduced via a parameter $h \in [0, 1]$, restricting cross-class imitation. Mutation occurs with probability μ.

The complete formula that takes into account the wealth classes is:

$$T_k^{X\to Y} = \frac{i_k^X}{Z}\left((1-\mu)\left[\frac{i_k^Y}{Z_k - 1 + (1-h)Z_l}\left(1 + e^{\beta(f_k^X - f_k^Y)}\right)^{-1} + \frac{(1-h)i_l^Y}{Z_k - 1 + (1-h)Z_l}\left(1 + e^{\beta(f_k^X - f_l^Y)}\right)^{-1}\right] + \mu\right)$$

Where $T_k^{X\to Y}$ is the probability of an agent using strategy X and of wealth class k to switch to a strategy Y.

The resulting process is a Markov chain over the state space (i_R, i_P).

4.3 Transition Matrix Construction

For each state (i_R, i_P), four elementary transitions are possible:

$$(i_R \pm 1, i_P), \quad (i_R, i_P \pm 1).$$

The transition matrix W is constructed entry-wise, with diagonal entries defined as:

$$W_{ii} = 1 - \sum_{j \neq i} W_{ji}.$$

4.4 Stationary Distribution

The stationary distribution π satisfies:

$$W\pi = \pi.$$

We compute π as the dominant eigenvector of W and normalize it to unit sum. This gives us the stationary distributions.

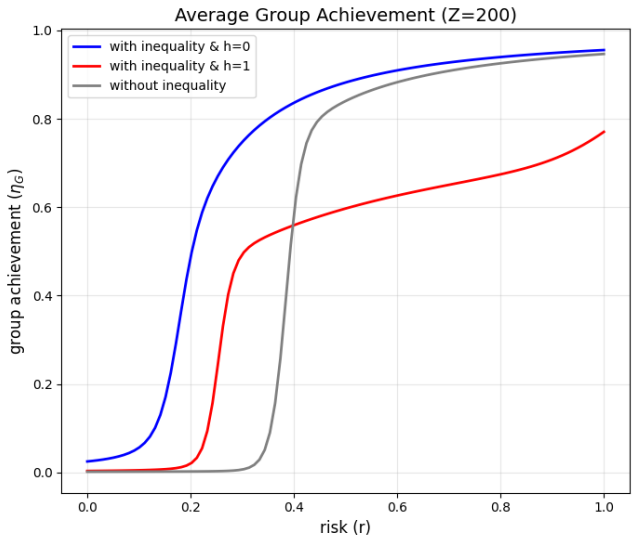


Figure 1: Average group achievement η_G as a function of perceived risk r.

4.5 Numerical Implementation

The full model is implemented in Python. Fitness values are computed by explicit enumeration of all admissible group compositions. The transition matrix is stored in sparse format.

```
def solve_stationary_distribution(self):
    W_dense = self.create_transition_matrix()
    W_sparse = sparse.csc_matrix(W_dense)
    evals, evecs = linalg.eigs(W_sparse, k=1,
                               sigma=1)
    stationary_dist = evecs[:, 0].real
    stationary_dist /= np.sum(stationary_dist)
    stationary_dist[stationary_dist < 0] = 0
    return stationary_dist
```

5 Results

5.1 Average Group Achievement

Figure 1 shows the average group achievement η_G as a function of risk r for three scenarios: wealth equality, wealth inequality with $h = 0$, and wealth inequality with $h = 1$.

This result shows that, as we said in the introduction, homophily has an higher impact than inequality in this threshold PGG.

5.2 Stationary Distributions

Figure 2 reproduces the stationary distributions and selection gradients for $(r, h) \in \{(0.2, 0), (0.2, 1), (0.3, 0), (0.3, 1)\}$.

The numerical configurations coincide with those reported by Vasconcelos et al. [1]. In particular, for $h = 0$, interior coexistence states dominate, whereas for $h = 1$ and low r, stationary distributions concentrate near full defection.

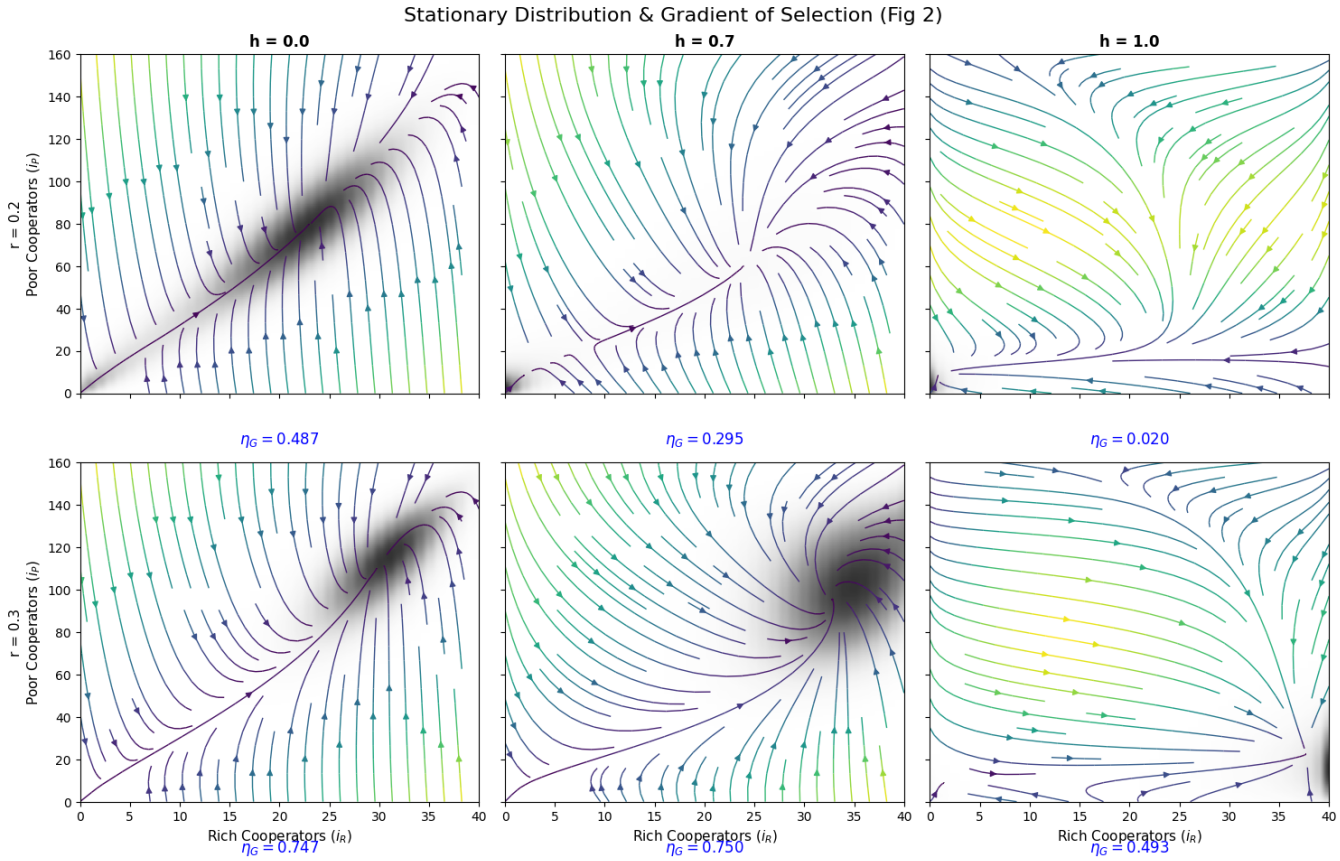


Figure 2: Stationary distributions and gradients of selection.

5.3 Effect of Obstinate Agents

Figure 3 illustrates the effects of obstinate behavior for $h = 1$ and $r = 0.2$.

Without obstinate agents, the system collapses into full defection, yielding $\eta_G \approx 0.003$. Introducing obstinate rich cooperators increases group achievement to $\eta_G \approx 0.206$, but cooperation remains asymmetric. In contrast, obstinate poor cooperators stabilize interior coexistence and yield $\eta_G \approx 0.583$.

6 Discussion

Our replication confirms the central qualitative claims of Vasconcelos et al. [1]. Wealth inequality alone does not destabilize cooperation; instead, homophily in imitation dynamics is the primary driver of collapse.

The asymmetry between rich and poor subpopulations plays a decisive role. While obstinate cooperation among rich agents improves outcomes, it fails to trigger system-wide cooperation. By contrast, persistent cooperation among poorer agents creates a stable contribution floor that induces reciprocal responses among wealthier individuals.

This finding challenges standard intuitions according to which influence scales monotonically with economic power. In collective-risk dilemmas, the strategic behavior of low-contributing agents may be structurally more important than that of high contributors.

Methodologically, our use of exact stationary distributions avoids sampling error and provides a precise characterization of long-run dynamics. This approach could be fruitfully applied to other evolutionary public goods models with small or structured populations.

7 Conclusion

This project presented an independent replication of the evolutionary model of climate cooperation under wealth inequality proposed by Vasconcelos et al. [1]. Using an exact finite-state Markov chain formulation, we reproduced the original phase diagrams, stationary distributions, and achievement curves.

Our findings reinforce the conclusion that homophily, rather than inequality per se, is the primary destabilizing force in collective-risk dilemmas. Moreover, we confirm the disproportionate stabilizing role of persistent cooperation among poorer agents.

These results suggest that climate policy frameworks should not only focus on aggregate contributions but also on the structure of social learning and imitation. Interventions that reduce segregation

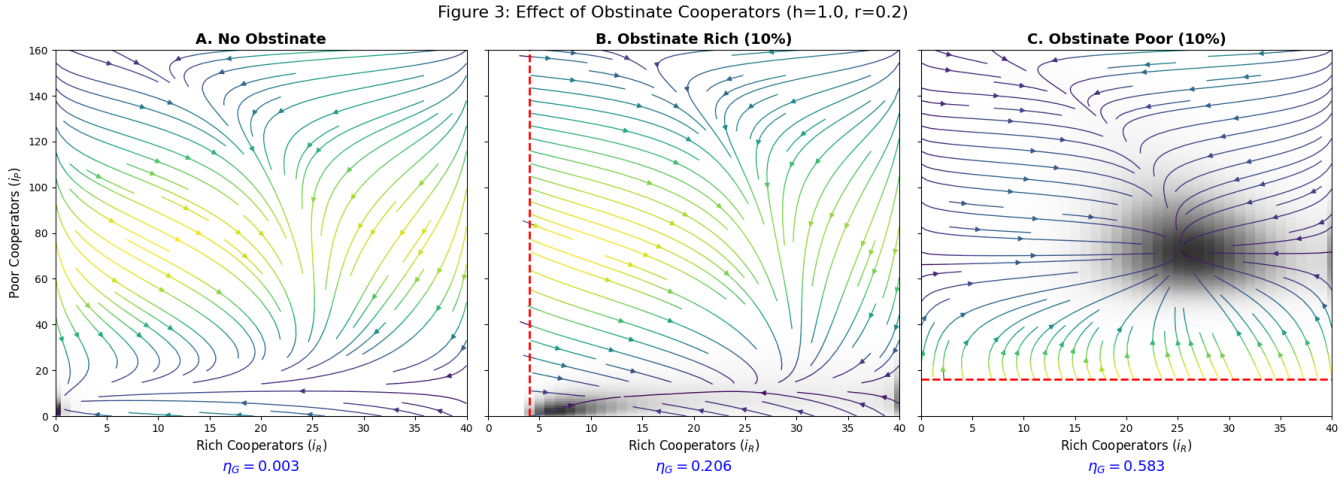


Figure 3: Effect of obstinate behavior: (A) no obstinate agents, (B) obstinate rich cooperators, (C) obstinate poor cooperators.

in information flow may be more effective than purely redistributive measures.

AI Usage

To facilitate visualization, we used generative AI tools to assist in drafting plotting scripts. All simulation logic, transition matrices, and numerical solvers were implemented independently. All AI-generated code was manually verified and adapted.

References

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